

MUHAMMAD FAISAL

Electrical Engineer | BMS & Automation | Embedded Systems | Solar & Power Systems

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 LinkedIn: [Muhammad Faisal](#) | Portfolio: [My Portfolio](#) | Iqama: Transferable

PROFESSIONAL SUMMARY

Results-driven Electrical Engineer with 5+ years of progressive experience in Building Management Systems (BMS), industrial automation, and embedded systems across Saudi Arabia's premier infrastructure projects including NEOM, KAFD, and Riyadh Metro. Proven expertise in Siemens Desigo CC, SCADA integration, Modbus/TCP/IP, energy optimization, and IoT-enabled monitoring solutions with hands-on testing and commissioning experience. Successfully developed and delivered Pakistan's first EV charging infrastructure and solar-BMS integration for NEOM's renewable energy initiative. Fluent in English with working knowledge of Arabic. Committed to delivering Saudi Vision 2030-aligned smart city and sustainability solutions for LEED and green building initiatives. Iqama transferable, available immediately.

- **Core Competencies:**
 - BMS Integration (Siemens Desigo CC, Xworks, ABT site, TIA portal, DXR/PXC, TXM)
 - Industrial Protocols (Modbus, RS485, CAN Bus, BACnet, KNX, IIOT, MQTT)
 - Embedded Systems (Atmel, STM32, ESP32, C/C++, Python, Linux)
 - PCB Design (Altium, Eagle - up to 6-layer)
 - SCADA Integration | Solar PV & BMS | HMI/GUI Development

PROFESSIONAL EXPERIENCE

SIEMENS | Riyadh, Saudi Arabia

Electrical Engineer | BMS | Automation | Commissioning | Nov 2023 – May 2026

- **Granada Business Park** | Jun 2025 – May 2026
 - Led BMS upgrade for 11 commercial buildings, replacing 105+ controllers & Field devices & integration with EMS system for improved fault detection by 40%
 - Commissioned DXR/PXC controllers and field devices; reduced maintenance response time by 50%
- **Riyadh Metro Line 3 - Power Shelters** | Oct 2025 – Jan 2026
 - Integrated Allen Bradley SCADA with Siemens BMS across 5+ power shelters;

reduced manual intervention by 65% and energy consumption by 18%

- Implemented Modbus RTU/TCP for sensors achieving 99.5% data accuracy; developed SCADA auto/manual control interfaces

- **Saudi National Bank HQ - KAFD** | Apr 2024 – May 2025

- Led BMS Upgradation & operations for 41-story high-rise building, managed 13-person team ensuring 100% on-time delivery
- Programming and commissioning of Siemens DXR/PXC controllers for AHU, FCU, FAHU, Chiller plants; upgraded legacy systems using DXR, PXC, QMX3, TXM modules & Desigo CC, increased reliability by 35%
- Provided 24/7 technical support maintaining 99.9% system uptime

- **NEOM – Solar, FACP BMS Integration** | Nov 2023 – Mar 2024

- Designed custom electronics and firmware for integration of solar inverters with Siemens DDC via Modbus TCP/IP
- Commissioned real-time solar monitoring, Fire Alarm integration and Robot Shelter integration across 13 NEOM zones; achieved 100% data transmission success

TESLA INDUSTRIES | Islamabad, Pakistan

Electrical Engineer - Embedded Systems | Jan 2020 – Oct 2023

- **EVX AC Charger:** Led end-to-end Hardware & firmware design of Pakistan's first 7kW/22kW EV charger with app control, dynamic power adjustment, and web interface; achieved 97.5% efficiency
- **EZ-Swap Stations:** Designed modular charging docks and hardware/firmware for Pakistan's first E-Bike sharing system using Modbus RTU and IOT via MQTT
- **30kW DC Power Supply:** Built HVDC lab supply with Nextion GUI and $\pm 0.5\%$ voltage regulation
- **Zero Export Manager:** Engineered 4-layer PCB control device preventing reverse grid feed-in; eliminated utility penalties for clients
- **Solar BMS Integration:** Programmed lithium BMS data integration via RS485 into solar inverter UI for SOC & other Vital parameters monitoring

EDUCATION

- **BSc in Electrical Engineering & Technology** | University of Sargodha, Pakistan | Sep 2018 – Oct 2022
- **DAE in Electronics** | HIRA College of Technology, Pakistan | Aug 2015 – Aug 2018

KEY ACHIEVEMENTS

- **NEOM Solar Integration:** Developed Electronics circuits to Integrate solar inverter data into Siemens BMS across 13 NEOM sites for real-time renewable monitoring
- **First EV Charger in Pakistan:** Designed Pakistan's first 7kW/22kW EV AC charger with smart scheduling and app control
- **30kW DC Power Source:** Developed GUI and firmware for high-voltage lab power supply with $\pm 0.5\%$ regulation
- **Zero Export Control:** Engineered control boards preventing reverse grid feed-in for solar installations